```
745
*************************************************************************
********************
745! *************************
746
747
*************************************************************************
********************
747! *************************
748
749
*************************************************************************
********************
749! *************************
750  20May09 When subdividing activity codes into pseudo codes to the
CARMM program, these are the
750!  places where they also
751                  will need to be added:
752                          Below where %addActvCodeToMxMail is being
called in (appends the
752! psuedo code to the mix mail map)
753                          Below where Proc format is formatting all
the activity codes
754                          In the excel file (ActivityCodesMaster
03AUG09.xls)
755                          In the sas code of "mapclassRecast.sas"
NOTE not needed for delivery
755!  cost model, as not
756                  using DMM field.
757              In the mixed mail/direct actv code match file
"MxMailCode.csv" - add new activity
757!  codes for Std Mail
758              Be sure to use the correct field in ActivityCodeMaster -
NewCRAProd
759                                  select KL.*, ac.NewCRAProd as class2
760
*************************************************************************
********************
760! ************************
761  REMOVE COMMENT ON SELECT CASING DEFINITION SECTION TO USE FOR CASING
ONLY!!!
762
*************************************************************************
********************
762! ************************;
763  resetline;
1    options MPRINT SYMBOLGEN;
2
3    libname IOCS  'D:\ExternalSSD\Folder 19\FY25\SAS\SASLib';
NOTE: Libref IOCS was successfully assigned as follows:
      Engine:        V9
      Physical Name: D:\ExternalSSD\Folder 19\FY25\SAS\SASLib
4    %let pathProg = D:\ExternalSSD\Folder 19\FY25\SAS\SASInputFiles;
5    %let pathOut =  D:\ExternalSSD\Folder 19\FY25\SAS\KLTables;
```

```
%let FY = 25;
%let FYData = 25;

%macro assignRouteGroup();
    ***  The following statement assigns route code groups
    ***  so that reports can be created separately for
    ***  these groups.;

        *set rgroup for special runs, for delivery cost model (3
values);
        *Put Letter Route types in RGroup 1, SPR in RGroup 2, 99 in
RGroup 3;

    if '71' <= route <= '83' then RGroup=1;
    else if '84' <= route <= '98' then RGroup=2;
    else RGroup=3;

%mend;


 data b;
 set IOCS.PRCPubCL25;
 title1 'FY 20&FY. Qtr A CARMM System';
  /*********** The following variables are updated using Cluster
dataset
**********************************/
 /****************  Updated by Dan Zhang on Nov 16, 2020
***********************************************/
    dollar = READCOST;   /*  Cost (in dollar) for each carrier
reading based on cluster
sampling method  */
           act1st = ' ';
       act234 = '   ';
    F9252 = CRAFTGRP;    /*  Craft Subscript: 5=Full-Time Carrier;
6=Part-Time Carrier  */
    f264 = WEIGHTCAG;    /*  CAG Group used in the Control Total
Dollar: A-H  */
/*********************************************************************
*******************
***************/


   ***  This secion is used for Folder 19, Delivery Cost Model
                   ****;
   ***  Subdivide ECR HD/SAT into DAL and non-DAL for use in Folder
19 Delivery Cost Model**;
   IF F262 in ('1317', '2317', '3317', '4317') then do;
      act1st = substr(F262,1,1);
      act234 = substr(F262,2,3);
      if  Q23E10    = 'N'  then act234 = '319'; ** Parent pc w/ no
DAL;
         else
      if  (Q23J02    = 'B'  or
```

```
45             Q23H03    = 'H') then do; ** WSH or MMS=EH marking eff
FY25Q1;
46             /*(Q23J02    = 'A'  or
47             Q23H03    = 'H') then do; ** WSH or MMS=EH marking; */
48
49             if (q23B01    = 'Y' and substr(F262,1,1) ne '1')
50                then act234 = '315'; ** DAL, for non-letters;
51             else act234 = '317';
52             end;
53            else
54         if (Q23J02    = 'A' or
55             Q23J02    = 'C' or
56             Q23H03    = 'I') then do; **WSS or MMS=ES marking  eff
FY25Q1;
57             /*(Q23J02    = 'B'  or
58             Q23H03    = 'I') then do; **WSS or MMS=ES marking; */
59
60             if (Q23B01    = 'Y'  and substr(F262,1,1) ne '1')
61             then act234 = '316'; ** DAL, for non-letters;
62             else act234 = '318';
63             end;
64         F262 = act1st||act234;
65      END;
66
67           ** CARMM Edits;
68           if f264='K' then f9252='9';
69           if f261 ne '4';
70           if dollar=0 and q01=' ' then delete;
71           else do;
72                 if f260 in ('84', '89') then f260='87';
73           end;
74           if f9252 in ('5','6'); **carrier craft;
75    run;

NOTE: Character values have been converted to numeric values at the
places given by:
      (Line):(Column).
      68:32   74:12
NOTE: Variable q01 is uninitialized.
NOTE: There were 150303 observations read from the data set
IOCS.PRCPUBCL25.
NOTE: The data set WORK.B has 102200 observations and 303 variables.
NOTE: DATA statement used (Total process time):
      real time           0.15 seconds
      cpu time            0.06 seconds


SYMBOLGEN:  Macro variable PATHPROG resolves to D:\ExternalSSD\Folder
19\FY25\SAS\SASInputFiles
76
77   *Read in file that maps activity codes into mixed mail codes,
78           and add all new activity codes added above;
79   %include "&pathProg\macMxMail_AMS.sas" / source2; /*GET MORE DETAILS
IN THE LOG FOR THIS
```

```
79 ! INCLUDE STATEMENT*/
NOTE: %INCLUDE (level 1) file D:\ExternalSSD\Folder
19\FY25\SAS\SASInputFiles\macMxMail_AMS.sas
      is file D:\ExternalSSD\Folder
19\FY25\SAS\SASInputFiles\macMxMail_AMS.sas.
80  +*Macros to read mixed mail activity code mapping, and insert new
activity codes;
81  +
82  +%macro readMxMail(CSVFile,MxMailMap);
83  +*Read file CSVFile (CSV format) with activity codes that belong to
mixed mail codes;
84  +*  Prepare final mixed mail mapping dataset;
85  +Data mxmail1;
86  + infile &CSVfile DLM="," MISSOVER firstobs=2;
87  + format mixcode $4.;
88  + format dircode $4.;
89  + input mixcode $ dircode $;
90  +run;
91  +
92  +Data &MxMailMap; set MxMail1;
93  +run;
94  +%mend;
95  +
96  +%macro addActvCodeToMxMail(MxMailMap, actvMatch, actvNew);
97  +*insert new activity code actvNew with same mixed mail cosdes as
actvMatch;
98  +proc sql;
99  + insert into MxMailMap
100 + select mixcode, &actvNew
101 + from mxmail1
102 + where dircode = &actvMatch;
103 +quit;
104 +%mend;
105 +
106 +
107 +*Inserts activity code subsets that Nancy needs for processing,
these are not official
107!+activity codes;
108 +*AMS 17Nov17;
109 +%macro addActvCodeToMaster(ActivityCodesTable, ActivityCode,
NewCRAProd);
110 +*insert new activity code actvNew with same mixed mail cosdes as
actvMatch;
111 +proc sql;
112 + insert into ActivityCodesTable
113 + set ActivityCode = &ActivityCode.,
114 +     NewCRAProd = &NewCRAProd.;
115 +quit;
116 +%mend;
117 +
118 +
119 +/*
120 +%macro
addMixedMailShapeToMxMail(MxMailMailMap,mxMailNoShape,shapeCode);
```

```
+*Insert new mixed mail code with shape info, using first digit of activity code
+ as last digit of mixed mail code;
+proc sql;
+  insert into MxMailMap
+    select put(&mxMailNoShape. + &shapeCode.,z4.), dirCode
+ from mxMail1
+ where mixCode = put(&mxMailNoShape,z4.) and substr(dirCode,1,1) = put(&shapeCode.,z1.);
+quit;
+%mend;*/
+%macro addMixedMailShapeToMxMail(MxMailMailMap,mxMailNoShape);
+*Insert new mixed mail code with shape info, using first digit of activity code
+ as last digit of mixed mail code;
+%let shapeCode = 1;
+proc sql;
+  insert into MxMailMap
+    select put(&mxMailNoShape. + &shapeCode.,z4.), dirCode
+ from mxMail1
+ where mixCode = put(&mxMailNoShape,z4.) and substr(dirCode,1,1) = put(&shapeCode.,z1.);
+%let shapeCode = 2;
+proc sql;
+  insert into MxMailMap
+    select put(&mxMailNoShape. + &shapeCode.,z4.), dirCode
+ from mxMail1
+ where mixCode = put(&mxMailNoShape,z4.) and substr(dirCode,1,1) = put(&shapeCode.,z1.);
+%let shapeCode = 3;
+proc sql;
+  insert into MxMailMap
+    select put(&mxMailNoShape. + &shapeCode.,z4.), dirCode
+ from mxMail1
+ where mixCode = put(&mxMailNoShape,z4.) and substr(dirCode,1,1) = put(&shapeCode.,z1.);
+%let shapeCode = 4;
+proc sql;
+  insert into MxMailMap
+    select put(&mxMailNoShape. + &shapeCode.,z4.), dirCode
+ from mxMail1
+ where mixCode = put(&mxMailNoShape,z4.) and substr(dirCode,1,1) = put(&shapeCode.,z1.);
+quit;
+%mend;
+
+
+/*
+*test macros;
+%let pathData = C:\AraSardar\IOCS\CARMM;
+%readMxMail("&pathData\MxMailCodeFY14.csv",MxMailMap)
+%addActvCodeToMxMail(MxMailMap,'1060','1061');
+%addActvCodeToMxMail(MxMailMap,'1060','1062');
+%addActvCodeToMxMail(MxMailMap,'1060','1063');
```

```
168 +%addActvCodeToMxMail(MxMailMap,'1310','1311');
169 +%addActvCodeToMxMail(MxMailMap,'1310','1312');
170 +%addActvCodeToMxMail(MxMailMap,'1310','1313');
171 +%addActvCodeToMxMail(MxMailMap,'1310','1314');
172 +%addActvCodeToMxMail(MxMailMap,'1310','1315');
173 +%addActvCodeToMxMail(MxMailMap,'1310','1316');
174 +%addActvCodeToMxMail(MxMailMap,'1310','1317');
175 +%addActvCodeToMxMail(MxMailMap,'1310','1318');
176 +%addActvCodeToMxMail(MxMailMap,'1310','1319');
177 +%addActvCodeToMxMail(MxMailMap,'2310','2311');
178 +%addActvCodeToMxMail(MxMailMap,'2310','2312');
179 +%addActvCodeToMxMail(MxMailMap,'2310','2313');
180 +%addActvCodeToMxMail(MxMailMap,'2310','2314');
181 +%addActvCodeToMxMail(MxMailMap,'2310','2315');
182 +%addActvCodeToMxMail(MxMailMap,'2310','2316');
183 +%addActvCodeToMxMail(MxMailMap,'2310','2317');
184 +%addActvCodeToMxMail(MxMailMap,'2310','2318');
185 +%addActvCodeToMxMail(MxMailMap,'2310','2319');
186 +%addActvCodeToMxMail(MxMailMap,'3310','3311');
187 +%addActvCodeToMxMail(MxMailMap,'3310','3312');
188 +%addActvCodeToMxMail(MxMailMap,'3310','3313');
189 +%addActvCodeToMxMail(MxMailMap,'3310','3314');
190 +%addActvCodeToMxMail(MxMailMap,'3310','3315');
191 +%addActvCodeToMxMail(MxMailMap,'3310','3316');
192 +%addActvCodeToMxMail(MxMailMap,'3310','3317');
193 +%addActvCodeToMxMail(MxMailMap,'3310','3318');
194 +%addActvCodeToMxMail(MxMailMap,'3310','3319');
195 +%addActvCodeToMxMail(MxMailMap,'4310','4311');
196 +%addActvCodeToMxMail(MxMailMap,'4310','4312');
197 +%addActvCodeToMxMail(MxMailMap,'4310','4313');
198 +%addActvCodeToMxMail(MxMailMap,'4310','4314');
199 +%addActvCodeToMxMail(MxMailMap,'4310','4315');
200 +%addActvCodeToMxMail(MxMailMap,'4310','4316');
201 +%addActvCodeToMxMail(MxMailMap,'4310','4317');
202 +%addActvCodeToMxMail(MxMailMap,'4310','4318');
203 +%addActvCodeToMxMail(MxMailMap,'4310','4319');
204 +proc sort data=MxMailMap nodup;
205 +by mixcode dircode;
206 +run;
207 +*/
208 +
209 +******* End macMxMail;
NOTE: %INCLUDE (level 1) ending.
SYMBOLGEN:  Macro variable PATHPROG resolves to D:\ExternalSSD\Folder
19\FY25\SAS\SASInputFiles
MPRINT(READMXMAIL):   *Read file CSVFile (CSV format) with activity codes
that belong to mixed
mail codes;
MPRINT(READMXMAIL):   * Prepare final mixed mail mapping dataset;
210  %readMxMail("&pathProg\MxMailCodeFY25.csv",MxMailMap);  *for
domestic CRA;
MPRINT(READMXMAIL):   Data mxmail1;
SYMBOLGEN:  Macro variable CSVFILE resolves to "D:\ExternalSSD\Folder
            19\FY25\SAS\SASInputFiles\MxMailCodeFY25.csv"
```

```
MPRINT(READMXMAIL):   infile "D:\ExternalSSD\Folder
19\FY25\SAS\SASInputFiles\MxMailCodeFY25.csv"
DLM="," MISSOVER firstobs=2;
MPRINT(READMXMAIL):   format mixcode $4.;
MPRINT(READMXMAIL):   format dircode $4.;
MPRINT(READMXMAIL):   input mixcode $ dircode $;
MPRINT(READMXMAIL):   run;

NOTE: The infile "D:\ExternalSSD\Folder
19\FY25\SAS\SASInputFiles\MxMailCodeFY25.csv" is:
      Filename=D:\ExternalSSD\Folder
19\FY25\SAS\SASInputFiles\MxMailCodeFY25.csv,
      RECFM=V,LRECL=32767,File Size (bytes)=2624,
      Last Modified=13Nov2025:12:44:54,
      Create Time=17Nov2025:10:11:34

NOTE: 237 records were read from the infile "D:\ExternalSSD\Folder
      19\FY25\SAS\SASInputFiles\MxMailCodeFY25.csv".
      The minimum record length was 9.
      The maximum record length was 9.
NOTE: The data set WORK.MXMAIL1 has 237 observations and 2 variables.
NOTE: DATA statement used (Total process time):
      real time           0.01 seconds
      cpu time            0.00 seconds


SYMBOLGEN:  Macro variable MXMAILMAP resolves to MxMailMap
MPRINT(READMXMAIL):   Data MxMailMap;
MPRINT(READMXMAIL):   set MxMail1;
MPRINT(READMXMAIL):   run;

NOTE: There were 237 observations read from the data set WORK.MXMAIL1.
NOTE: The data set WORK.MXMAILMAP has 237 observations and 2 variables.
NOTE: DATA statement used (Total process time):
      real time           0.00 seconds
      cpu time            0.00 seconds


211
MPRINT(ADDACTVCODETOMXMAIL):   *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
212
213
214  %addActvCodeToMxMail(MxMailMap,'1317','1315');
MPRINT(ADDACTVCODETOMXMAIL):   proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '1315'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '1317'
MPRINT(ADDACTVCODETOMXMAIL):   insert into MxMailMap select mixcode,
'1315' from mxmail1 where
dircode = '1317';
NOTE: 4 rows were inserted into WORK.MXMAILMAP.

MPRINT(ADDACTVCODETOMXMAIL):   quit;
```

```
NOTE: PROCEDURE SQL used (Total process time):
      real time           0.00 seconds
      cpu time            0.00 seconds


MPRINT(ADDACTVCODETOMXMAIL):   *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
215  %addActvCodeToMxMail(MxMailMap,'1317','1316');
MPRINT(ADDACTVCODETOMXMAIL):   proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '1316'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '1317'
MPRINT(ADDACTVCODETOMXMAIL):   insert into MxMailMap select mixcode,
'1316' from mxmail1 where
dircode = '1317';
NOTE: 4 rows were inserted into WORK.MXMAILMAP.

MPRINT(ADDACTVCODETOMXMAIL):   quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time           0.00 seconds
      cpu time            0.00 seconds


MPRINT(ADDACTVCODETOMXMAIL):   *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
216  %addActvCodeToMxMail(MxMailMap,'1317','1318');
MPRINT(ADDACTVCODETOMXMAIL):   proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '1318'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '1317'
MPRINT(ADDACTVCODETOMXMAIL):   insert into MxMailMap select mixcode,
'1318' from mxmail1 where
dircode = '1317';
NOTE: 4 rows were inserted into WORK.MXMAILMAP.

MPRINT(ADDACTVCODETOMXMAIL):   quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time           0.00 seconds
      cpu time            0.00 seconds


MPRINT(ADDACTVCODETOMXMAIL):   *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
217  %addActvCodeToMxMail(MxMailMap,'1317','1319');
MPRINT(ADDACTVCODETOMXMAIL):   proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '1319'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '1317'
MPRINT(ADDACTVCODETOMXMAIL):   insert into MxMailMap select mixcode,
'1319' from mxmail1 where
dircode = '1317';
NOTE: 4 rows were inserted into WORK.MXMAILMAP.

MPRINT(ADDACTVCODETOMXMAIL):   quit;
```

```
NOTE: PROCEDURE SQL used (Total process time):
      real time           0.01 seconds
      cpu time            0.00 seconds


MPRINT(ADDACTVCODETOMXMAIL):   *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
218  %addActvCodeToMxMail(MxMailMap,'2317','2315');
MPRINT(ADDACTVCODETOMXMAIL):   proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '2315'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '2317'
MPRINT(ADDACTVCODETOMXMAIL):   insert into MxMailMap select mixcode,
'2315' from mxmail1 where
dircode = '2317';
NOTE: 4 rows were inserted into WORK.MXMAILMAP.

MPRINT(ADDACTVCODETOMXMAIL):   quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time           0.00 seconds
      cpu time            0.00 seconds


MPRINT(ADDACTVCODETOMXMAIL):   *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
219  %addActvCodeToMxMail(MxMailMap,'2317','2316');
MPRINT(ADDACTVCODETOMXMAIL):   proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '2316'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '2317'
MPRINT(ADDACTVCODETOMXMAIL):   insert into MxMailMap select mixcode,
'2316' from mxmail1 where
dircode = '2317';
NOTE: 4 rows were inserted into WORK.MXMAILMAP.

MPRINT(ADDACTVCODETOMXMAIL):   quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time           0.00 seconds
      cpu time            0.00 seconds


MPRINT(ADDACTVCODETOMXMAIL):   *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
220  %addActvCodeToMxMail(MxMailMap,'2317','2318');
MPRINT(ADDACTVCODETOMXMAIL):   proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '2318'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '2317'
MPRINT(ADDACTVCODETOMXMAIL):   insert into MxMailMap select mixcode,
'2318' from mxmail1 where
dircode = '2317';
NOTE: 4 rows were inserted into WORK.MXMAILMAP.

MPRINT(ADDACTVCODETOMXMAIL):   quit;
```

```
NOTE: PROCEDURE SQL used (Total process time):
      real time           0.00 seconds
      cpu time            0.00 seconds


MPRINT(ADDACTVCODETOMXMAIL):   *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
221  %addActvCodeToMxMail(MxMailMap,'2317','2319');
MPRINT(ADDACTVCODETOMXMAIL):   proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '2319'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '2317'
MPRINT(ADDACTVCODETOMXMAIL):   insert into MxMailMap select mixcode,
'2319' from mxmail1 where
dircode = '2317';
NOTE: 4 rows were inserted into WORK.MXMAILMAP.

MPRINT(ADDACTVCODETOMXMAIL):   quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time           0.00 seconds
      cpu time            0.00 seconds


MPRINT(ADDACTVCODETOMXMAIL):   *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
222  %addActvCodeToMxMail(MxMailMap,'3317','3315');
MPRINT(ADDACTVCODETOMXMAIL):   proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '3315'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '3317'
MPRINT(ADDACTVCODETOMXMAIL):   insert into MxMailMap select mixcode,
'3315' from mxmail1 where
dircode = '3317';
NOTE: 3 rows were inserted into WORK.MXMAILMAP.

MPRINT(ADDACTVCODETOMXMAIL):   quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time           0.00 seconds
      cpu time            0.00 seconds


MPRINT(ADDACTVCODETOMXMAIL):   *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
223  %addActvCodeToMxMail(MxMailMap,'3317','3316');
MPRINT(ADDACTVCODETOMXMAIL):   proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '3316'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '3317'
MPRINT(ADDACTVCODETOMXMAIL):   insert into MxMailMap select mixcode,
'3316' from mxmail1 where
dircode = '3317';
NOTE: 3 rows were inserted into WORK.MXMAILMAP.

MPRINT(ADDACTVCODETOMXMAIL):   quit;
```

```
NOTE: PROCEDURE SQL used (Total process time):
      real time           0.00 seconds
      cpu time            0.00 seconds


MPRINT(ADDACTVCODETOMXMAIL):   *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
224  %addActvCodeToMxMail(MxMailMap,'3317','3318');
MPRINT(ADDACTVCODETOMXMAIL):   proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '3318'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '3317'
MPRINT(ADDACTVCODETOMXMAIL):   insert into MxMailMap select mixcode,
'3318' from mxmail1 where
dircode = '3317';
NOTE: 3 rows were inserted into WORK.MXMAILMAP.

MPRINT(ADDACTVCODETOMXMAIL):   quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time           0.00 seconds
      cpu time            0.00 seconds


MPRINT(ADDACTVCODETOMXMAIL):   *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
225  %addActvCodeToMxMail(MxMailMap,'3317','3319');
MPRINT(ADDACTVCODETOMXMAIL):   proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '3319'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '3317'
MPRINT(ADDACTVCODETOMXMAIL):   insert into MxMailMap select mixcode,
'3319' from mxmail1 where
dircode = '3317';
NOTE: 3 rows were inserted into WORK.MXMAILMAP.

MPRINT(ADDACTVCODETOMXMAIL):   quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time           0.01 seconds
      cpu time            0.00 seconds


MPRINT(ADDACTVCODETOMXMAIL):   *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
226  %addActvCodeToMxMail(MxMailMap,'4317','4315');
MPRINT(ADDACTVCODETOMXMAIL):   proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '4315'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '4317'
MPRINT(ADDACTVCODETOMXMAIL):   insert into MxMailMap select mixcode,
'4315' from mxmail1 where
dircode = '4317';
NOTE: 3 rows were inserted into WORK.MXMAILMAP.

MPRINT(ADDACTVCODETOMXMAIL):   quit;
```

```
NOTE: PROCEDURE SQL used (Total process time):
      real time           0.01 seconds
      cpu time            0.00 seconds


MPRINT(ADDACTVCODETOMXMAIL):   *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
227  %addActvCodeToMxMail(MxMailMap,'4317','4316');
MPRINT(ADDACTVCODETOMXMAIL):   proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '4316'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '4317'
MPRINT(ADDACTVCODETOMXMAIL):   insert into MxMailMap select mixcode,
'4316' from mxmail1 where
dircode = '4317';
NOTE: 3 rows were inserted into WORK.MXMAILMAP.

MPRINT(ADDACTVCODETOMXMAIL):   quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time           0.00 seconds
      cpu time            0.00 seconds


MPRINT(ADDACTVCODETOMXMAIL):   *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
228  %addActvCodeToMxMail(MxMailMap,'4317','4318');
MPRINT(ADDACTVCODETOMXMAIL):   proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '4318'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '4317'
MPRINT(ADDACTVCODETOMXMAIL):   insert into MxMailMap select mixcode,
'4318' from mxmail1 where
dircode = '4317';
NOTE: 3 rows were inserted into WORK.MXMAILMAP.

MPRINT(ADDACTVCODETOMXMAIL):   quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time           0.00 seconds
      cpu time            0.00 seconds


MPRINT(ADDACTVCODETOMXMAIL):   *insert new activity code actvNew with
same mixed mail cosdes as
actvMatch;
229  %addActvCodeToMxMail(MxMailMap,'4317','4319');
MPRINT(ADDACTVCODETOMXMAIL):   proc sql;
SYMBOLGEN:  Macro variable ACTVNEW resolves to '4319'
SYMBOLGEN:  Macro variable ACTVMATCH resolves to '4317'
MPRINT(ADDACTVCODETOMXMAIL):   insert into MxMailMap select mixcode,
'4319' from mxmail1 where
dircode = '4317';
NOTE: 3 rows were inserted into WORK.MXMAILMAP.

MPRINT(ADDACTVCODETOMXMAIL):   quit;
```

```
NOTE: PROCEDURE SQL used (Total process time):
      real time           0.00 seconds
      cpu time            0.00 seconds


230
231
232  proc sort data=MxMailMap nodup;
233          by dircode mixcode;
234  run;

NOTE: There were 293 observations read from the data set WORK.MXMAILMAP.
NOTE: 0 duplicate observations were deleted.
NOTE: The data set WORK.MXMAILMAP has 293 observations and 2 variables.
NOTE: PROCEDURE SORT used (Total process time):
      real time           0.01 seconds
      cpu time            0.00 seconds


235
236  data carr; set b;
237    Craft=F9252;
238    Actv=F262;
239    *Actv=F9806; *mcz 10Aug08 use F9806 to distribute Int'l mixed
mail;
240    Route=F260;
241    BF=F261;
242    Fin=F263;
243    Cag=F264;
244    Cagfin=cag !! fin;
245
246    ****  SELECT CASING DEFINITION *** ;
247    *To restrict to subset that is casing remove comment from next
line;
248   if Q16F03A in ('A','B','C');  ** General casing (3 options) -- use
for Delivery Cost Model
248! inputs;
249
250     if actv in ('5740', '5745', '5741') then actv = '5750';  /*as per
Audrey Hu, 17Nov23*/
251     if actv <= '0900' and actv not in ('0350','0560') then actgrp=1;
252       else if '1020' <= actv <= '4910' or actv in ('0350','0560')
then actgrp=2;
253       else if '5300' <= actv <= '5750' then actgrp=3;
254       else if '5040' <= actv <= '6720' then actgrp=4;
SYMBOLGEN:  Macro variable FY resolves to 25
255    TITLE1 "FY 20&fy. Qtr A CARMM SYSTEM";
256  run;

NOTE: There were 102200 observations read from the data set WORK.B.
NOTE: The data set WORK.CARR has 26447 observations and 311 variables.
NOTE: DATA statement used (Total process time):
      real time           0.04 seconds
      cpu time            0.01 seconds
```

```

proc sql;
  create table smy as
  select cag, fin, actv, bf, sum(dollar) as dollar
  from carr
  group by cag, fin, actv, bf;
NOTE: Table WORK.SMY created, with 320 rows and 5 columns.

quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time           0.01 seconds
      cpu time            0.01 seconds



 data smyx;
    array bfx{4} bf1 - bf4;
    do i=1 to 4;
     set smy;
     by cag fin actv bf;
      if i=1 and bf='2' then i=i+1;
      if i=1 and bf='3' then i=i+2;
      if i=1 and bf='5' then i=i+3;
      if i=2 and bf='3' then i=i+1;
      if i=2 and bf='5' then i=i+2;
      if i=3 and bf='5' then i=i+1;
      bfx{i}=dollar;
      if last.actv then i=4;
    end;
     if bf1=. then bf1=0;
     if bf2=. then bf2=0;
     if bf3=. then bf3=0;
     if bf4=. then bf4=0;
     total=sum(of bf1-bf4);
     cagfin=cag !! fin;
     if actv <= '0900' and actv not in ('0350','0560') then
actgrp=1;
       else if '1020' <= actv <= '4910' or actv in ('0350','0560')
then actgrp=2;
       else if '5300' <= actv <= '5750' then actgrp=3;
       else if '5040' <= actv <= '6720' then actgrp=4;
        output;
 run;

NOTE: There were 320 observations read from the data set WORK.SMY.
NOTE: The data set WORK.SMYX has 320 observations and 13 variables.
NOTE: DATA statement used (Total process time):
      real time           0.01 seconds
      cpu time            0.00 seconds

```

```
 proc sort data=smyx;
       by cagfin actgrp actv;
 run;

NOTE: There were 320 observations read from the data set WORK.SMYX.
NOTE: The data set WORK.SMYX has 320 observations and 13 variables.
NOTE: PROCEDURE SORT used (Total process time):
      real time           0.00 seconds
      cpu time            0.00 seconds


    *----------------------------------------------------------;
    *                                                          ;
    * Dollar tally summary by cag fin route actv bf.  This     ;
    * summary file is used in next step for distributing       ;
    * mixed-mail.                                             ;
    *                                                          ;
    *----------------------------------------------------------;

proc sql;
  create table smy2 as
  select cag, fin, route, actv, bf, sum(dollar) as dollar
  from carr
  group by cag, fin, route, actv, bf;
NOTE: Table WORK.SMY2 created, with 1117 rows and 6 columns.

quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time           0.02 seconds
      cpu time            0.00 seconds


 data smy2;
    set smy2;

*Use macro to assign route group;
        %assignRouteGroup;
MPRINT(ASSIGNROUTEGROUP):   *** The following statement assigns route
code groups *** so that
reports can be created separately for *** these groups.;
MPRINT(ASSIGNROUTEGROUP):   *set rgroup for special runs, for delivery
cost model (3 values);
MPRINT(ASSIGNROUTEGROUP):   *Put Letter Route types in RGroup 1, SPR in
RGroup 2, 99 in RGroup 3;
MPRINT(ASSIGNROUTEGROUP):   if '71' <= route <= '83' then RGroup=1;
MPRINT(ASSIGNROUTEGROUP):   else if '84' <= route <= '98' then RGroup=2;
MPRINT(ASSIGNROUTEGROUP):   else RGroup=3;
    cagfin=cag !! fin;
 run;

NOTE: There were 1117 observations read from the data set WORK.SMY2.
```

```
NOTE: The data set WORK.SMY2 has 1117 observations and 8 variables.
NOTE: DATA statement used (Total process time):
      real time           0.01 seconds
      cpu time            0.00 seconds



data out;
       array bff{4} dollar1-dollar4;
       do i=1 to 4;
        set smy2;
        by cag fin route actv bf;
         if i=1 and bf='2' then i=i+1;
         if i=1 and bf='3' then i=i+2;
         if i=1 and bf='5' then i=i+3;
         if i=2 and bf='3' then i=i+1;
         if i=2 and bf='5' then i=i+2;
         if i=3 and bf='5' then i=i+1;
         bff{i}=dollar;
         if last.actv then i=4;
       end;
        if dollar1=. then dollar1=0;
        if dollar2=. then dollar2=0;
        if dollar3=. then dollar3=0;
        if dollar4=. then dollar4=0;
        drop i cag fin _type_ _freq_ dollar bf;
       output;
run;

WARNING: The variable _type_ in the DROP, KEEP, or RENAME list has never
been referenced.
WARNING: The variable _freq_ in the DROP, KEEP, or RENAME list has never
been referenced.
NOTE: There were 1117 observations read from the data set WORK.SMY2.
NOTE: The data set WORK.OUT has 1117 observations and 8 variables.
NOTE: DATA statement used (Total process time):
      real time           0.01 seconds
      cpu time            0.00 seconds




  *************************************************************;
  ****Note: this 2nd   part of the program is analogous to the **;
  ****      old program ALA860C0, which performed distribution **;
  ****      and produced various summary reports.              **;
  *************************************************************;

data out;
     set out;
     if '5300' <= actv <= '5750' then mixmail=1;
     else if '1000' <= actv <= '5000' or actv in ('0350','0560')
then mixmail=0;
     dollart=sum(of dollar1 - dollar4);
```

```
run;

NOTE: There were 1117 observations read from the data set WORK.OUT.
NOTE: The data set WORK.OUT has 1117 observations and 10 variables.
NOTE: DATA statement used (Total process time):
      real time           0.00 seconds
      cpu time            0.00 seconds



   *-------------------------------------------------------------;
   * The following proc SUMMARY gets the preliminary results for  ;
   * Report 7  This corresponds to the old LIOCATT report ALA860P7;
   *-------------------------------------------------------------;
proc sort data=out out=out7;
      by rgroup route actv;
      where mixmail ne 1;
run;

NOTE: There were 938 observations read from the data set WORK.OUT.
      WHERE mixmail not = 1;
NOTE: The data set WORK.OUT7 has 938 observations and 10 variables.
NOTE: PROCEDURE SORT used (Total process time):
      real time           0.01 seconds
      cpu time            0.00 seconds



proc  summary data=out7 noprint;
      var dollar1 dollar2 dollar3 dollar4;
      output out=rpt7 sum=;
      by rgroup route actv;
    /****uncomment print if report is needed***
    proc print data=rpt7;
      id rgroup route ;
    * sum dollar1 dollar2 dollar3 dollar4;
      by rgroup route;
      title2 'Report 7';
     *****************************************/
run;

NOTE: There were 938 observations read from the data set WORK.OUT7.
NOTE: The data set WORK.RPT7 has 227 observations and 9 variables.
NOTE: PROCEDURE SUMMARY used (Total process time):
      real time           0.00 seconds
      cpu time            0.00 seconds



   **************************************************************;
   * Distribute mixed mail costs to direct mail activity codes.   ;
   **************************************************************;

   *-------------------------------------------------------------;
```

```
     * Read the mixed-mail to direct mail code conversion file.     ;
     * This file tells which direct mail activity codes to associate;
     * with each mixed-mail activity code.                           ;
     *--------------------------------------------------------------;
         ;
Data mxmail;
set MxMailMap;
run;

NOTE: There were 293 observations read from the data set WORK.MXMAILMAP.
NOTE: The data set WORK.MXMAIL has 293 observations and 2 variables.
NOTE: DATA statement used (Total process time):
      real time           0.01 seconds
      cpu time            0.00 seconds



 proc sort data=mxmail; by dircode;
 run;

NOTE: There were 293 observations read from the data set WORK.MXMAIL.
NOTE: The data set WORK.MXMAIL has 293 observations and 2 variables.
NOTE: PROCEDURE SORT used (Total process time):
      real time           0.01 seconds
      cpu time            0.00 seconds



 data dirmail (drop= i mixcode);
       array dir(8) $mix1 -mix8;
       do i=1 to 8;
          set mxmail;
          by dircode;
          actv=dircode;
          dir(i)=mixcode;
          if last.dircode then i=8;
       end;
       output;
 run;

NOTE: There were 293 observations read from the data set WORK.MXMAIL.
NOTE: The data set WORK.DIRMAIL has 101 observations and 10 variables.
NOTE: DATA statement used (Total process time):
      real time           0.01 seconds
      cpu time            0.00 seconds


     *--------------------------------------------------------------;
     * Direct mail cost summary by cagfin, route and activity       ;
     *--------------------------------------------------------------;
 proc sort data=out out=dist2;
         by cagfin route actv;
         where mixmail=0; * Contains only direct mail;
 run;
```

```
NOTE: There were 890 observations read from the data set WORK.OUT.
      WHERE mixmail=0;
NOTE: The data set WORK.DIST2 has 890 observations and 10 variables.
NOTE: PROCEDURE SORT used (Total process time):
      real time           0.01 seconds
      cpu time            0.00 seconds


proc summary data=dist2 noprint;
       var dollar1 dollar2 dollar3 dollar4 dollart;
       output out=dirdol
              sum=dollar1d dollar2d dollar3d dollar4d dollartd;
        by cagfin route actv;
       title2 'dirdol';
   * proc print data=dirdol;
run;

NOTE: There were 890 observations read from the data set WORK.DIST2.
NOTE: The data set WORK.DIRDOL has 890 observations and 10 variables.
NOTE: PROCEDURE SUMMARY used (Total process time):
      real time           0.01 seconds
      cpu time            0.00 seconds


   *-------------------------------------------------------------;
   * Merge direct mail cost with the conversion table to associate;
   * each direct mail activity code with mixed mail code that will;
   * get distribued cost.                                        ;
   *-------------------------------------------------------------;
proc sort data=dirdol;
       by actv;
               run;

NOTE: There were 890 observations read from the data set WORK.DIRDOL.
NOTE: The data set WORK.DIRDOL has 890 observations and 10 variables.
NOTE: PROCEDURE SORT used (Total process time):
      real time           0.01 seconds
      cpu time            0.00 seconds


proc sort data=dirmail;
       by actv;
run;

NOTE: There were 101 observations read from the data set WORK.DIRMAIL.
NOTE: The data set WORK.DIRMAIL has 101 observations and 10 variables.
NOTE: PROCEDURE SORT used (Total process time):
      real time           0.00 seconds
      cpu time            0.00 seconds
```

```
data dircode1 (drop=mix1-mix8 i);
        merge dirdol  (in=a)
              dirmail (in=b keep=actv mix1-mix8);
        by actv;
        source=a+2*b;
        if source=2 then delete; * delete activity codes that
contain
                                   no cost;
        array dir(8) mix1-mix8;
        do i=1 to 8;
          mixcode=dir(i);
          if mixcode ne ' ' then output;
          if mixcode= ' ' then i=8;
         end;
   *proc print n;
run;

NOTE: There were 890 observations read from the data set WORK.DIRDOL.
NOTE: There were 101 observations read from the data set WORK.DIRMAIL.
NOTE: The data set WORK.DIRCODE1 has 2868 observations and 12 variables.
NOTE: DATA statement used (Total process time):
      real time           0.01 seconds
      cpu time            0.00 seconds



   *-------------------------------------------------------------;
   * Mixed mail cost summary by cagfin, route and actv           ;
   *-------------------------------------------------------------;

proc sort data=out out=dist1;
        by cagfin route actv;
        where mixmail=1; * Contains only mixed mails;
                run;

NOTE: There were 179 observations read from the data set WORK.OUT.
      WHERE mixmail=1;
NOTE: The data set WORK.DIST1 has 179 observations and 10 variables.
NOTE: PROCEDURE SORT used (Total process time):
      real time           0.00 seconds
      cpu time            0.00 seconds



proc summary data=dist1 noprint;
       var dollar1 dollar2 dollar3 dollar4 dollart;
       output out=mxdol
              sum=dollar1i dollar2i dollar3i dollar4i dollarti;
        by cagfin route actv;
       title2 'mxdol';
       run;

NOTE: There were 179 observations read from the data set WORK.DIST1.
```

```
NOTE: The data set WORK.MXDOL has 179 observations and 10 variables.
NOTE: PROCEDURE SUMMARY used (Total process time):
      real time           0.01 seconds
      cpu time            0.00 seconds



     *--------------------------------------------------------------;
     * Attach summary mixed mail cost to the direct mail activity   ;
     * code.                                                        ;
     *--------------------------------------------------------------;

data mxdol;
       set mxdol;
       rename actv=mixcode;
    *proc print data=mxdol;
        run;

NOTE: There were 179 observations read from the data set WORK.MXDOL.
NOTE: The data set WORK.MXDOL has 179 observations and 10 variables.
NOTE: DATA statement used (Total process time):
      real time           0.00 seconds
      cpu time            0.00 seconds



proc sort data=dircode1;
        by cagfin route mixcode;
                run;

NOTE: There were 2868 observations read from the data set WORK.DIRCODE1.
NOTE: The data set WORK.DIRCODE1 has 2868 observations and 12 variables.
NOTE: PROCEDURE SORT used (Total process time):
      real time           0.00 seconds
      cpu time            0.00 seconds


proc sort data=mxdol;
        by cagfin route mixcode;
                run;

NOTE: There were 179 observations read from the data set WORK.MXDOL.
NOTE: The data set WORK.MXDOL has 179 observations and 10 variables.
NOTE: PROCEDURE SORT used (Total process time):
      real time           0.00 seconds
      cpu time            0.00 seconds



data final ;
        merge dircode1 (in=a) mxdol (in=b);
        by cagfin route mixcode;
        source=a+2*b;

```

```
      * proc print;
      * title2 'final';
        run;

NOTE: There were 2868 observations read from the data set WORK.DIRCODE1.
NOTE: There were 179 observations read from the data set WORK.MXDOL.
NOTE: The data set WORK.FINAL has 2868 observations and 17 variables.
NOTE: DATA statement used (Total process time):
      real time           0.00 seconds
      cpu time            0.00 seconds



     *--------------------------------------------------------------;
     * Calculate the total direct mail cost associated with         ;
     * each mixed mail code.                                        ;
     *--------------------------------------------------------------;

proc sort data=final;
        by cagfin route mixcode;
                run;

NOTE: There were 2868 observations read from the data set WORK.FINAL.
NOTE: The data set WORK.FINAL has 2868 observations and 17 variables.
NOTE: PROCEDURE SORT used (Total process time):
      real time           0.01 seconds
      cpu time            0.00 seconds



 proc summary data=final noprint;
       var dollar1d dollar2d dollar3d dollar4d dollartd;
        output out=total1
               sum=total1d total2d total3d total4d totaltd;
        by cagfin route mixcode;
        run;

NOTE: There were 2868 observations read from the data set WORK.FINAL.
NOTE: The data set WORK.TOTAL1 has 410 observations and 10 variables.
NOTE: PROCEDURE SUMMARY used (Total process time):
      real time           0.01 seconds
      cpu time            0.00 seconds



     *--------------------------------------------------------------;
     * Distribute the mixed mail cost to direct mail activity codes ;
     *--------------------------------------------------------------;
        data distfile (drop=_type_ _freq_ source);
           merge final (in=a) total1 (in=b);
              by cagfin route mixcode;

           if total1d=0 then ratio1=.;
              else ratio1= dollar1d/total1d;
```

```
          dist_1=dollar1i *ratio1;

          if total2d=0 then ratio2=.;
             else ratio2= dollar2d/total2d;
          dist_2=dollar2i *ratio2;

          if total3d=0 then ratio3=.;
             else ratio3= dollar3d/total3d;
          dist_3=dollar3i *ratio3;

     ***  The following statements add the undistributed dollar
     ***  for basic functions 1 - 3 to last basic function;

          if (total1d =0 and dollar1i > 0)
             then dollar4i=dollar4i +dollar1i;
          if (total2d =0 and dollar2i > 0)
             then dollar4i=dollar4i +dollar2i;
          if (total3d =0 and dollar3i > 0)
             then dollar4i=dollar4i +dollar3i;

     ***  Note that the last basic function distribution is based on
     ***  the total cost (sum of four basic functions cost);

          if totaltd=0 then ratiot=.;
          else ratiot= dollartd/totaltd;
          dist_4=dollar4i *ratiot;

     ***  The following code assigns the undistributed mixed
     ***  dollars to the dist_0 cagegory;

        if actv=' ' then
           do;
              actv=mixcode;
              dist_0=dollarti;
           end;

     *Use macro to assign route group;
     %assignRouteGroup;
MPRINT(ASSIGNROUTEGROUP):   *** The following statement assigns route
code groups *** so that
reports can be created separately for *** these groups.;
MPRINT(ASSIGNROUTEGROUP):   *set rgroup for special runs, for delivery
cost model (3 values);
MPRINT(ASSIGNROUTEGROUP):   *Put Letter Route types in RGroup 1, SPR in
RGroup 2, 99 in RGroup 3;
MPRINT(ASSIGNROUTEGROUP):   if '71' <= route <= '83' then RGroup=1;
MPRINT(ASSIGNROUTEGROUP):   else if '84' <= route <= '98' then RGroup=2;
MPRINT(ASSIGNROUTEGROUP):   else RGroup=3;


     * proc print data=distfile n;
 run;
```

```
NOTE: Missing values were generated as a result of performing an
operation on missing values.
      Each place is given by: (Number of times) at (Line):(Column).
      2868 at 520:27   745 at 524:27    2868 at 528:27   745 at 545:27
NOTE: There were 2868 observations read from the data set WORK.FINAL.
NOTE: There were 410 observations read from the data set WORK.TOTAL1.
NOTE: The data set WORK.DISTFILE has 2868 observations and 29 variables.
NOTE: DATA statement used (Total process time):
      real time           0.03 seconds
      cpu time            0.00 seconds


562
563           *-------------------------------------------------------------;
564           * Combine schedule K and L to produce K and L report         ;
565           * (report 13). Schedule K is report 7 and schedule L is      ;
566           * summary of the mixed mail distribution result (finout).   ;
567           *-------------------------------------------------------------;
568
569     proc sort data=distfile;
570            by  rgroup route actv;
571                    run;

NOTE: There were 2868 observations read from the data set WORK.DISTFILE.
NOTE: The data set WORK.DISTFILE has 2868 observations and 29 variables.
NOTE: PROCEDURE SORT used (Total process time):
      real time           0.01 seconds
      cpu time            0.00 seconds


572
573     proc summary data=distfile noprint;
574           var  dist_0 dist_1 dist_2 dist_3 dist_4;
575           output out=finoutl sum=dollar0 dollar1 dollar2 dollar3
dollar4;
576            by  rgroup route actv;
577           run;

NOTE: There were 2868 observations read from the data set WORK.DISTFILE.
NOTE: The data set WORK.FINOUTL has 205 observations and 10 variables.
NOTE: PROCEDURE SUMMARY used (Total process time):
      real time           0.01 seconds
      cpu time            0.00 seconds


578
579    *-------------------------------------------------------------;
580    *  The following proc format classifies activity codes into ;
581    *  categories for subsequent reports.                        ;
582    *-------------------------------------------------------------;
583    proc format ;
584    value $actv
585    /*SPECIAL SERVICES  */
586           '0060'       ='055 Registered  '
```

```
          '0050'       ='051 Certified   '
          '0080'       ='054 Insured     '
          '0030'       ='052 COD         '

    /* OTHSERV INCL. BUSRPLY, RETRCPT, ADDRCORR, FORMS3547&3579 */
          '6020','6030'='074 PO Box    '
          '0090','0190',
          '0300','0100',
          '0110','0140'              ='058 Other Services'

    /* FIRST CLASS   */
          '1060'                     ='003 FC SP Letters  '
          '1020'                     ='004 FC SP Cards    '
          '1080','1081','1085','1086'='008 FC Prst Letters'
          '1022','1035','1040','1045'='009 FC Prst Cards  '
          '2060'                     ='014 FC SP Flats    '
          '2080','2081','2086'       ='017 FC Prst Flats  '
          '3060','4060'              ='143 FC Package Svc '
          /*'3080','4080'                ='020 FC Prst Parcels'*/
          '3080','4080'              ='143 FC Package Svc '


    /*PRIORITY MAIL  */
          '1160','2160','3160','4160'='148 Priority       '

    /* EXPRESS MAIL    */
             '1110','2110','3110','4110' ='140 Express    '

    /* PERIODICALS       */
             '1211','2211','3211','4211'='031 PER In County     '
             '1212','2212','3212','4212'='032 PER Outside County'

    /* STANDARD          */
             '1317'                     ='021 MM ECR HD Letters
'
             '2317','3317','4317'       ='022 MM ECR HD Flats/Parcels
'
             '1318'                     ='021 MM ECR SAT Letters
'
             '2318','3318','4318'       ='022 MM ECR SAT
Flats/Parcels  '
             '1312','2312','3312','4312'='023 MM ECR Carrier Route
'
             '1320','2320','3320','4320'='024 MM EDDM - Retail
'
             '1340','1341','1345'       ='025 MM REG Letters
'
             '2340','2341','2345'       ='026 MM REG Flats
'
             '3340','4340'              ='027 MM REG Parcel
'
             '3341','4341'              ='027 MM REG Parcel
'
```

```
              '1315'                     ='021 DAL Ltr WSH '
              '2315','3315','4315'       ='021 DAL Flt/IPP/PAR WSH '
              '1316'                     ='022 DAL Ltr WSS '
              '2316','3316','4316'       ='022 DAL Flt/IPP/PAR WSS '
              '1319','2319','3319','4319'='021 Parent No Dal HD/SAT '

     /* PACKAGE          */
             '2402','3402','4402'        ='157 Ground Advantage HW 1-3lb '
             '2403','3403','4403'        ='158 Ground Advantage HW over3lb '
             '2405','3405','4405'        ='156 Ground Advantage LW '
             '2410','3410','4410'        ='145 PKG Retail Ground '
             '2460','2465','2480','2495' ='042 PKG BPM Flats '
             '3460','3465','3480','3495',
             '4460','4465','4480','4495' ='043 PKG BPM Parcels '
             '1420','1425','1430',
             '2420','2425','2430',
             '3420','3425','3430',
             '4420','4425','4430'        ='044 PKG Media '

     /* PARCEL SELECT and PS Lightweight        */
             '2360','3360','4360',
             '2493','3493','4493'        ='151 PKG Parcel Select '

     /* PARCEL RETURN SERVICE (PRS)          */
             '2475','3475','4475'        ='154 PKG Parcel Return (PRS) '

    /* PREMIUM FORWARDING SERVICE (PFS)     */
              '0350','1170','2170','3170','4170'='170 Prem. Forw. Svc'


    /* GOVERNMENT MAIL   */
              '0560',
              '1510','2510','3510','4510'='085 USPS       '

     /* FREE               */
              '1910','2910','3910','4910'='086 Free       '

     /* INTERNATIONAL OUTBOUND AND INBOUND  */
              '1780','2780','3780','4780',
              '5081','6081','5082','6082',
```

```
                '0752','0757','0762','0767',
                '0770','0860','0862','0867'              ='185
International'

/* OTHER CODES  */
                OTHER = '999 OTHER';
NOTE: Format $ACTV is already on the library WORK.FORMATS.
NOTE: Format $ACTV has been output.
run;

NOTE: PROCEDURE FORMAT used (Total process time):
      real time           0.03 seconds
      cpu time            0.00 seconds



data KL (drop=_type_ _freq_);
        set rpt7 (in=a) finoutl (in=b);
        if a then source='K';
        if b then source='L';
        if dollar0 > 0 then
           do;   source='U';
                 dollar4=dollar0; *move the undistributed mixed
mail
                                   cost to basic function OTHER;
           end;

        if dollar0=. and dollar1=. and dollar2=. and dollar3=. and
         dollar4=. then delete;

        total= sum (of dollar1 - dollar4);
                if actv <= '0900' and actv not in ('0350','0560')
then actgrp=1;
               else if '1020' <= actv <= '4910' or actv in
('0350','0560') then actgrp=2;
           else if '5300' <= actv <= '5750' then actgrp=3;
           else if '5020' <= actv <= '6720' then actgrp=4;**bls;
        **use format to classify activity to classes***;
        class=put(actv,$actv.);

        **assign labels for use in reports**;
        label
           dollar1 = 'OUTGOING'
           dollar2 = 'INCOMING'
           dollar3 = 'TRANSIT '
           dollar4 = 'OTHER   ';

                     **restrict to letter routes only**;
                     *if rgroup=1;
run;

NOTE: There were 227 observations read from the data set WORK.RPT7.
NOTE: There were 205 observations read from the data set WORK.FINOUTL.
```

```
NOTE: The data set WORK.KL has 430 observations and 12 variables.
NOTE: DATA statement used (Total process time):
      real time           0.03 seconds
      cpu time            0.00 seconds



*Generate class variable from lookup table;
*import craft,activity code combinations used in delivery cost
model;
PROC IMPORT OUT= WORK.ActivityCodesTable
            DATAFILE=
"&pathprog\ActivityCodesMaster_FilingFY25.xlsx"
SYMBOLGEN:  Macro variable PATHPROG resolves to D:\ExternalSSD\Folder
19\FY25\SAS\SASInputFiles
            DBMS=XLSX REPLACE;
RUN;

NOTE: One or more variables were converted because the data type is not
supported by the V9
      engine. For more details, run with options MSGLEVEL=I.
NOTE: The import data set has 163 observations and 6 variables.
NOTE: WORK.ACTIVITYCODESTABLE data set was successfully created.
NOTE: PROCEDURE IMPORT used (Total process time):
      real time           0.02 seconds
      cpu time            0.00 seconds


proc sql;
create table KL2 as
  select KL.*, ac.NewCRAProd as class2
  from KL left join ActivityCodesTable as ac
  on KL.actv = ac.ActivityCode;
NOTE: Table WORK.KL2 created, with 430 rows and 13 columns.

quit;
NOTE: PROCEDURE SQL used (Total process time):
      real time           0.02 seconds
      cpu time            0.00 seconds



data checkClass; set KL2;
where class ne class2;
run;

NOTE: There were 120 observations read from the data set WORK.KL2.
      WHERE class not = class2;
NOTE: The data set WORK.CHECKCLASS has 120 observations and 13 variables.
NOTE: DATA statement used (Total process time):
      real time           0.00 seconds
      cpu time            0.00 seconds
```

```
***zzz;

data KL; set KL2;
if class2 = "" then class2 = "999 OTHER";
if substr(class2,1,3) = "185" then class = "185 International";
        else class = class2;
drop class2;
run;

NOTE: There were 430 observations read from the data set WORK.KL2.
NOTE: The data set WORK.KL has 430 observations and 12 variables.
NOTE: DATA statement used (Total process time):
      real time           0.00 seconds
      cpu time            0.00 seconds


proc sort data=kl;
by rgroup route actv;
run;

NOTE: There were 430 observations read from the data set WORK.KL.
NOTE: The data set WORK.KL has 430 observations and 12 variables.
NOTE: PROCEDURE SORT used (Total process time):
      real time           0.01 seconds
      cpu time            0.00 seconds


*Export KL table by route, for delivery cost model;
proc export data=kl
    outfile = "&pathOut\KLTable_Casing&FYData..xlsx"
SYMBOLGEN:  Macro variable PATHOUT resolves to D:\ExternalSSD\Folder
19\FY25\SAS\KLTables
SYMBOLGEN:  Macro variable FYDATA resolves to 25
        dbms = xlsx replace;
run;

NOTE: The export data set has 430 observations and 12 variables.
NOTE: "D:\ExternalSSD\Folder 19\FY25\SAS\KLTables\KLTable_Casing25.xlsx"
file was successfully
      created.
NOTE: PROCEDURE EXPORT used (Total process time):
      real time           0.06 seconds
      cpu time            0.00 seconds
```